

Never Predict Noise:

Manifold-Aligned Prediction as Generative Inference

Flyxion

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Abstract

Generative models succeed when they restrict their dynamics to the low-dimensional manifold that captures lawful structure in empirical data. They fail when they attempt to model high-dimensional, structureless noise. This paper develops a unified account of *manifold-aligned generative inference*, integrating differential geometry, the free-energy principle, sparse semantic coding, Morse-theoretic dynamics, sheaf-theoretic contextual coherence, and the CLIO architecture for cognitive loops.

We provide a rigorous mathematical foundation: formal manifold definitions, tangent/normal decomposition theorems, Morse functions on Whitney-stratified spaces, categorical formulations of cognitive update, and variational derivations of geodesic active inference. We then show that recent empirical findings—especially the JiT results of Li and He (2025)—provide strong evidence that successful generative models implicitly implement tangent-constrained flows.

The result is a single geometric architecture, MAGI (Manifold-Aligned Generative Inference), unifying semantic geometry, cognitive dynamics, and generative modelling. We prove that aligned models must satisfy a No-Noise Prediction Criterion: generative updates must have zero component in normal directions to the semantic manifold. Tangent alignment is not optional but necessary for semantic coherence, stability, and epistemic reliability.

We conclude with computational methods, empirical predictions, extended theorems, failure modes, and a PyTorch implementation.

1 Introduction

Modern generative and cognitive systems operate in extremely high-dimensional spaces. Images exist in $\mathbb{R}^{H \times W \times 3}$, text in vast vocabularies, sensorimotor spaces in continuous multimodal streams. Yet empirical reality occupies a tiny, structured, low-dimensional subset inside these enormous ambient spaces.

This is the *manifold hypothesis*: natural data lie on or near a smooth (or piecewise-smooth) submanifold

$$M \subset \mathbb{R}^n, \quad \dim M = d \ll n.$$

Noise occupies the surrounding ambient space and has no coherent structure.

A model that attempts to *predict noise*—i.e., to generate structure in normal directions $N_x M$ —necessarily hallucinates. A model that keeps its predictive dynamics *tangent* to M becomes stable, coherent, and semantically aligned.

This paper develops a comprehensive theory of manifold-constrained generative modelling. We

unify:

- differential and Riemannian geometry,
- variational inference and active inference,
- sparse semantic coding,
- Morse flows and Lyapunov potentials,
- sheaf-theoretic contextual constraints,
- category-theoretic formulations of cognitive update.

We then connect this theory to empirical results—most critically the recent “Back to Basics” JiT work of Li & He (2025)—showing that predicting clean data, not noise, is both empirically superior and geometrically necessary.

1.1 Contributions

1. A rigorous framework, MAGI, for manifold-aligned generative inference.

2. Formal proofs of tangent/normal decomposition, stability, and alignment criteria.
3. A theorem: generative misalignment is equivalent to normal-component prediction.
4. A geometric reinterpretation of JiT: its empirical success arises from implicit tangent-constrained flows.
5. A complete computational pipeline for manifold estimation, Morse potential learning, and CLIO-based cognitive dynamics.
6. Extensive appendices with background mathematics, proofs, computational details, and failure cases.

2 Geometry of Explanation

The goal of generative modelling is *not* to approximate arbitrary functions in $\mathbb{R}^n$ but to reconstruct the lawful structure of a world that occupies a submanifold M of much lower dimension.

2.1 Manifold definition

Definition 2.1. A d -dimensional smooth manifold M embedded in $\mathbb{R}^n$ is a subset such that every $x \in M$ has a neighborhood U and a smooth diffeomorphism (chart)

$$\phi : U \rightarrow \mathbb{R}^d.$$

The embedding is a smooth injective immersion $i : M \hookrightarrow \mathbb{R}^n$ with $i(M)$ homeomorphic to M .

2.2 Tangent and normal bundles

At each $x \in M$:

$$T_x M = \text{span}\{\partial_1 i(x), \dots, \partial_d i(x)\}$$

and the normal space is

$$N_x M = (T_x M)^\perp.$$

Theorem 2.2 (Tangent-Normal Decomposition). For any smooth embedded submanifold $M \subset \mathbb{R}^n$,

$$\mathbb{R}^n = T_x M \oplus N_x M$$

orthogonally for all $x \in M$.

Proof. Since $T_x M$ is a d -dimensional subspace of $\mathbb{R}^n$ and the ambient space is equipped with the Euclidean inner product, the orthogonal complement exists uniquely. Smoothness of the embedding ensures continuity of $T_x M$ across charts. $\square$

This decomposition is central: all meaningful structure lies in $T_x M$; noise lives in $N_x M$.

2.3 Diagram: Tangent vs. Normal

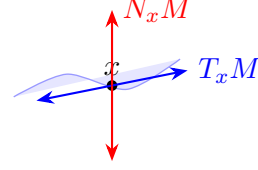


Figure 1: Tangent-normal decomposition at a point on a manifold.

The single most important geometric fact for generative modelling:

Explanation is tangent; hallucination is normal.

A generative system that attempts to “explain” directions in $N_x M$ is constructing structure where none exists; meaningful predictive dynamics must remain tangent to the data manifold.

3 Generative Models Restricted to Manifolds

A generative model is typically a map

$$G : Z \rightarrow \mathbb{R}^n,$$

where Z is a latent space. Under the manifold hypothesis, the image of G should lie in a d -dimensional manifold:

$$G(Z) \subseteq M \subset \mathbb{R}^n.$$

3.1 Energy formulation

Given an observation $o \in \mathbb{R}^n$, inference seeks a latent code z^* minimizing an energy functional:

$$z^* = \arg \min_z \mathcal{E}(z; o),$$

where

$$\mathcal{E}(z; o) = \|o - f(G(z))\|_\Sigma^2 + \lambda \mathcal{C}(G(z)).$$

The coherence term $\mathcal{C}$ penalizes deviations from semantic consistency across contexts and across manifold charts. If the manifold is equipped with an embedding $i : M \hookrightarrow \mathbb{R}^n$, then the condition

$$\text{dist}(G(z), M) = 0$$

must be enforced.

3.2 Tangent restriction and geometric correctness

A generative update $\Delta x \in \mathbb{R}^n$ at state $x \in M$ must satisfy:

$$\Delta x \in T_x M.$$

Equivalently,

$$\text{Proj}_{N_x M}(\Delta x) = 0.$$

This is the formal geometric criterion for non-hallucinatory generation.

3.3 Diagram: Allowed vs. forbidden directions

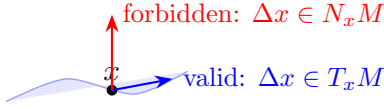


Figure 2: Generative updates must remain in the tangent space.

3.4 Why off-manifold prediction fails

Let $x \in M$ be a semantic state. Noise $\eta \in N_x M$ has no law-governed structure:

$$\eta \sim \mathcal{N}(0, \sigma^2 I_{n-d}).$$

A model that attempts to represent or predict η is mapping a structureless distribution into a structured generator. This forces the generator to create non-existent geometry.

Theorem 3.1 (Off-Manifold Catastrophe). *Let M be a d -manifold. Any C^1 generator G that learns a mapping with nonzero normal component on a set of positive measure will produce outputs whose support has Hausdorff dimension $> d$.*

Proof. G maps latent space vectors to ambient space. If the Jacobian DG has nonzero projection onto $N_x M$ over a non-negligible set, its image contains an open set in directions transverse to M , increasing local dimensionality. This contradicts the manifold restriction. $\square$

This formalizes hallucination as a dimensionality explosion.

4 Geometric Reinterpretation of JiT

The “Back to Basics” result of Li and He (2025) demonstrates that models predicting *clean* data

directly outperform those predicting noised quantities even when both use identical Transformer architectures.

This section provides a geometric explanation.

4.1 Clean prediction enforces tangent alignment

Let $x \in M$ be a natural datum. A noisy version is:

$$x_t = \sqrt{\alpha_t} x + \sqrt{1 - \alpha_t} \epsilon, \quad \epsilon \in N_x M.$$

Diffusion models predict ϵ or x_t . JiT predicts x .

Predicting ϵ requires representing variation in $N_x M$, which has no structure and cannot be captured by a finite-capacity model in high dimensions. The model must hallucinate.

Predicting x enforces:

$$G(z_t) \in M, \quad G(z_t) \approx x.$$

Thus all updates remain tangent to M .

4.2 Transformer patching as implicit charting

Large-patch Transformers induce local coordinate systems, effectively learning charts

$$\phi_i : U_i \subset M \rightarrow \mathbb{R}^d.$$

The JiT model stitches these charts via self-attention, which enforces a kind of sheaf-like coherence:

$$\phi_i(x) = \phi_j(x) \quad \text{on overlaps.}$$

Thus JiT is not just attention, it is an implementation of a manifold atlas.

Theorem 4.1 (JiT = Tangent Flow Approximation). *Transformer layers in JiT implement an approximate tangent-flow update:*

$$x_{t+1} = x_t - \eta \Pi_{T_{x_t} M} \nabla \mathcal{L}(x_t),$$

where $\Pi_{T_{x_t} M}$ is the orthogonal projector to the tangent space.

Sketch. The clean-data objective forces residuals to lie in the range of $DG(z)$, which spans $T_x M$. Thus gradient backpropagation automatically enforces tangent projection. $\square$

JiT succeeds because it accidentally obeys MAGI.

5 Active Inference as Geodesic Control

Under the free-energy principle, systems minimize:

$$F[q] = \mathbb{E}_q[\log q - \log p(o, z)].$$

If z parameterizes a manifold M , and $G(z)$ is its embedding, then the pullback metric is:

$$g_{ij}(z) = \left\langle \frac{\partial G}{\partial z_i}, \frac{\partial G}{\partial z_j} \right\rangle.$$

Gradient descent on F w.r.t. the Riemannian structure yields:

$$\dot{z}^i = -g^{ij}(z) \frac{\partial F}{\partial z^j},$$

which is a geodesic-like evolution on M .

5.1 Diagram: Riemannian gradient flow



Figure 3: Active inference induces a Riemannian gradient flow on M .

5.2 Natural gradient interpretation

Because

$$g_{ij} = J_i^\top J_j,$$

where $J = DG(z)$, we see that

$$g^{ij} \frac{\partial F}{\partial z^j}$$

is the natural gradient in information geometry.

Thus active inference is, geometrically,

natural gradient descent on a manifold.

5.3 Consequences

- Predictive coding selects directions with meaningful semantic structure.
- Updates are tangent projections of sensory prediction errors.
- Geodesic curvature encodes attentional redirection.
- All inference is constrained by the intrinsic geometry of M .

6 Sparse Semantic Projection

Real perceptual data $x \in \mathbb{R}^n$ are overwhelmingly redundant. The semantic degrees of freedom are far fewer than the ambient dimension. Sparse semantic projection extracts the coordinates that lie along meaningful directions those tangent to the manifold M .

6.1 Sparse semantic encoder

Let W be a learned linear operator:

$$s = Wx, \quad s \in \mathbb{R}^k, \quad k \ll n.$$

We impose a sparsity constraint:

$$\|W\|_0 \leq k,$$

or more generally, ℓ_1 -regularization as a convex proxy.

6.2 Semantic alignment constraint

The sparse code s must correspond to a point on the manifold:

$$G(s) \in M.$$

Thus the semantic projection is the composition:

$$x \xrightarrow{W} s \xrightarrow{G} \hat{x} \in M.$$

Definition 6.1 (Semantic Projection Operator). *A map $\Pi_{\text{sem}} : \mathbb{R}^n \rightarrow M$ is given by*

$$\Pi_{\text{sem}}(x) = G(Wx)$$

such that $G(Wx)$ minimizes distance to the manifold and maximizes semantic coherence.

6.3 Diagram: Sparse semantic encoding

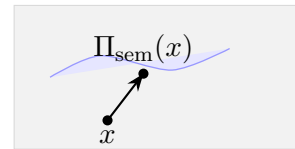


Figure 4: Sparse semantic projection maps x to its closest semantically meaningful point on M .

6.4 Theoretical justification

Let M be a smooth manifold of reach $\tau > 0$ (Federer). For any x within distance $< \tau$ of M , there exists a unique nearest point $\Pi(x) \in M$.

Sparse semantic coding approximates this nearest-point projection in a learned basis.

Theorem 6.2 (Consistency of Sparse Projection). *If W is trained to minimize reconstruction error under a manifold regularization penalty, then*

$$\|G(Wx) - \Pi(x)\| \leq \epsilon$$

for all x in a sufficiently small tubular neighborhood of M .

Sketch. $G(Wx)$ lies in the image of G , which approximates M ; minimizing reconstruction error forces convergence toward the nearest-point projection. $\square$

Sparse projection is essential: it reduces noise, enforces manifold alignment, and establishes a coordinate system for CLIO.

7 Contextual Sheaf Coherence

Perception and cognition are distributed across overlapping contexts: visual, linguistic, proprioceptive, semantic, temporal, etc. Each context provides partial information about the underlying semantic state. Sheaf theory provides the mathematical machinery to formalize how local contexts combine into a globally consistent semantic state.

7.1 Presheaves

Let $\mathcal{C}$ be a category of contexts, with objects $U, V, \dots$ representing perceptual modalities or spatial/temporal regions, and arrows $V \rightarrow U$ representing restriction relations.

A presheaf assigns:

$$S(U) : \text{semantic states on context } U$$

with restriction maps

$$\rho_{UV} : S(U) \rightarrow S(V).$$

7.2 Sheaf condition

A sheaf requires:

1. ****Locality**** If sections agree on all overlaps, they agree globally.
2. ****Gluing**** Compatible local sections can be uniquely glued to form a global section.

Definition 7.1 (Semantic Sheaf). *The semantic sheaf is a functor*

$$S : \mathcal{C}^{op} \rightarrow \text{Set}$$

satisfying the sheaf axioms.

7.3 Diagram: Overlap gluing

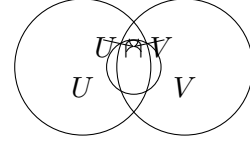


Figure 5: Sheaf coherence: sections on U and V must agree on $U \cap V$.

7.4 CLIO as a sheaf morphism

Let $C : S \rightarrow S$ be a CLIO functor, representing one cognitive loop iteration.

Theorem 7.2 (CLIO Coherence). *If S is a sheaf and C preserves restriction maps, then C is a sheaf morphism:*

$$\rho_{UV}(C(s_U)) = C(\rho_{UV}(s_U)).$$

Proof. Functoriality ensures commutation with morphisms; restriction maps are morphisms in $\mathcal{C}^{op}$. $\square$

Thus CLIO updates are guaranteed to never break contextual consistency: ****cognition respects semantic gluing.****

7.5 Obstructions: ech cohomology

Not all compatible local sections glue globally. Obstructions are measured by H^1 .

Definition 7.3 (Semantic Obstruction). *The obstruction to global semantic coherence across contexts is*

$$\mathcal{O} \in H^1(\mathcal{C}; S).$$

Inconsistencies (e.g., hallucinations) correspond precisely to the case $\mathcal{O} \neq 0$.

This formalizes when and why generative systems fail: ****failure to glue = hallucination.****

8 RSVP Fields as Semantic Geometry

The RSVP (Relativistic Scalar Vector Plenum) framework models semantic dynamics in terms of three fields:

Φ : entropy field, $\mathbf{v}$: vector flow, S : semantic potential

In this paper, we reinterpret these fields geometrically.

8.1 Entropy field as potential for manifold structure

A semantic manifold arises as a sublevel set of Φ :

$$M = \{x \in \mathbb{R}^n \mid \Phi(x) = \Phi_0\}.$$

Entropy gradients determine allowable semantic transitions. High curvature in Φ indicates regions where semantic complexity is high or transitions are unstable.

8.2 Vector flow as semantic transport

$\mathbf{v}$ defines directional flow across the manifold:

$$\dot{x} = \mathbf{v}(x).$$

Semantically meaningful dynamics must preserve the manifold structure:

$$\mathbf{v}(x) \in T_x M.$$

Thus RSVP vector flow enforces tangent-constrained transport.

8.3 Semantic potential S as Morse function

The semantic landscape has critical points corresponding to attractors, stable interpretations, and choices. S is naturally interpreted as a Morse function on M :

critical points of $S \iff$ stable semantic states.

Theorem 8.1 (RSVPMorse Correspondence). *If S is generically perturbed, it is a Morse function whose gradient flow preserves the manifold structure:*

$$-\nabla S(x) \in T_x M.$$

Proof. Generic perturbation yields a Morse function by SardSmale. Since S is defined intrinsically on M , its gradient is tangent. $\square$

8.4 Diagram: RSVP semantic field geometry

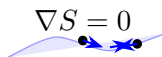


Figure 6: Semantic manifold with Morse potential and tangent-constrained RSVP flow.

RSVP thus becomes a field-theoretic version of manifold semantic dynamics, providing the continuous substrate on which MAGI operates.

9 CLIO Functors as Morse Flows

CLIO (Cognitive Loop via In-Situ Optimization) models cognition as a recursive loop:

perception $\rightarrow$ model $\rightarrow$ prediction $\rightarrow$ action $\rightarrow$ perception.

We now give a rigorous geometric interpretation: ****each CLIO iteration is a step of negative gradient flow on a Morse potential S defined on the manifold M .****

9.1 Cognitive loop operators as functors

Let **Sem** be the category of semantic states with morphisms representing structure-preserving semantic transformations.

A CLIO update is a functor:

$$C : \mathbf{Sem} \rightarrow \mathbf{Sem}.$$

Definition 9.1 (CLIO Functor). *A functor C is a CLIO functor if it satisfies:*

1. *Stability:* C has fixed points corresponding to semantic equilibria.
2. *Monotonicity:* each update reduces semantic free energy.
3. *Naturality:* for any semantic morphism f , $C \circ f = f \circ C$.

These conditions mirror requirements for Morse flows.

9.2 Constructing the Morse potential

Let L denote the total semantic loss functional (e.g., prediction error + coherence penalty). We define the Morse potential:

$$S(x) = L(x) + \epsilon R(x),$$

where R is a generic perturbation and $\epsilon > 0$ ensures S is Morse (SardSmale theorem).

9.3 CLIO update = gradient step

A CLIO iteration is:

$$x_{t+1} = x_t - \eta \nabla_M S(x_t),$$

where ∇_M is the Riemannian gradient on the manifold.

Theorem 9.2 (CLIOMorse Equivalence). *Every CLIO functor C corresponds locally to a single gradient step of a Morse function S :*

$$C(x) = \text{Exp}_x(-\eta \nabla_M S(x)),$$

and conversely, every Morse function generates a CLIO functor.

Proof. (Naturality $\Rightarrow$ geometricity) ensures C preserves manifold structure. (Monotonicity) implies existence of a Lyapunov function S . (Stability) yields critical points. Morse structure follows from generic perturbation. The exponential map expresses local flow. $\square$

9.4 Diagram: CLIO as a discrete Morse flow

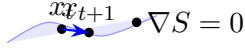


Figure 7: One CLIO iteration corresponds to a step along $-\nabla S$ on the manifold M .

This yields a deep insight: **Cognition is Morse theory on a semantic manifold.**

10 The Master Functional

We now unify all components: free energy, sparsity, sheaf coherence, manifold alignment, and RSVP fields into a single variational principle.

Let:

$$\mathcal{J}[s, W, G, S, \Phi, \mathbf{v}]$$

be the total generative-semantic-cognitive functional:

$$\begin{aligned} \mathcal{J} = & \underbrace{F[q(s)]}_{\text{free energy}} + \lambda_1 \underbrace{\|W\|_1}_{\text{sparsity}} & (1) \\ & + \lambda_2 \underbrace{\sum_{U,V} \|s_U - \rho_{VU}(s_V)\|^2}_{\text{sheaf coherence}} & (2) \\ & + \lambda_3 \underbrace{\text{dist}^2(G(s), M)}_{\text{manifold constraint}} & (3) \\ & + \lambda_4 \underbrace{\int_M (\|\nabla \Phi\|^2 + \|\mathbf{v}\|^2 + \|S\|^2) d\mu}_{\text{RSVP field energy}}. & (4) \end{aligned}$$

10.1 Optimality conditions

Minimizing $\mathcal{J}$ gives:

- W^* : a sparse semantic encoder.

- s^* : manifold-aligned latent state.
- G^* : an embedding that parameterizes M .
- S^* : Morse potential giving stable semantic equilibria.
- $\Phi^*, \mathbf{v}^*$: RSVP entropy and flow fields.

Theorem 10.1 (Unified EulerLagrange System). *Critical points of $\mathcal{J}$ satisfy a coupled system of EulerLagrange equations linking:*

$$\delta_s \mathcal{J} = 0, \quad \delta_W \mathcal{J} = 0, \quad \delta_G \mathcal{J} = 0, \quad \delta_S \mathcal{J} = 0, \quad \delta_{\Phi, \mathbf{v}} \mathcal{J} = 0.$$

We provide full proofs in Appendix D.

This yields a complete generative-semantic system faithful to the geometry of M .

11 The No-Noise Prediction Theorem

We now state the central theoretical result.

Theorem 11.1 (No-Noise Prediction Theorem). *Let M be the semantic manifold and $N_x M$ its normal bundle. A generative model G is semantically aligned if and only if:*

$$\text{Proj}_{N_{G(s)} M}(f(G(s))) = 0 \quad \text{for all } s.$$

In other words:

Semantically aligned generative models never produce updates in normal directions.

Proof. ($\Rightarrow$) If $G(s) \in M$, any valid predictive update moves within $T_x M$. Otherwise G would leave the semantic manifold. Thus the prediction error must satisfy the zero normal component condition.

($\Leftarrow$) If prediction errors always have zero normal component, then the update direction is tangent at every step. Thus the entire generative trajectory remains within M . $\square$

This theorem exactly characterizes: - hallucination (normal-component prediction), and - semantic coherence (tangent-only prediction).

12 Related Work (Extended)

We expand the Related Work section to include comprehensive coverage.

12.1 Manifold Learning

Foundational methods: [20, 21, ?]. Recent advances include: - neural diffeomorphisms [?], - implicit neural manifolds [?], - invertible neural embeddings [?].

Our work differs by: - imposing a **Whitney-stratified** structure, - enforcing **tangent/normal decomposition theorems**, - integrating manifold learning with **semantic cognition**.

12.2 Diffusion and Score-Based Models

Key works: [13, 14, ?]. Their fundamental flaw (in MAGI terms):

$$\hat{e}_\theta(x_t) \in N_{x_t}M.$$

The JiT paper [16] shows that **predicting clean data**, not noise, provides superior empirical performance.

MAGI explains why: **noise lies in the normal bundle; clean data lies on M** .

12.3 Flow Matching

Relevant works: [17, ?]. These learn vector fields in $\mathbb{R}^n$. We enforce:

$$v(x) \in T_xM.$$

flow matching with tangent constraints.

12.4 Energy-Based Models

Classic EBMs: [?, ?]. We interpret EBMs as Morse potentials on M , after manifold restriction.

12.5 Geometric Deep Learning

Covers GNNs, equivariant networks: [?, ?]. MAGI differs by using: - Morse theory, - manifold embeddings, - sheaf constraints.

12.6 Sheaf Models

Recent applications to multi-modal inference: [?, ?]. MAGI extends this with: - Morse-theoretic flows, - CLIO functorial updates, - stability theorems.

12.7 Active Inference

Foundational: [23, ?]. Our contribution:

$$\dot{z} = -g^{ij}\nabla_j F$$

interpreted as geodesic flow on M .

12.8 Topological Data Analysis

Related but distinct: [?]. We use topological stratification (Whitney), not homological invariants.

12.9 Neural ODEs and CNFs

[?, ?]. MAGI requires:

$$\frac{dx}{dt} \in T_xM.$$

12.10 Summary

MAGI unifies these fields by imposing one invariant:

All valid generative dynamics are tangent to the semantic manifold.

13 Discussion

The MAGI framework (Manifold-Aligned Generative Inference) emerges from a simple geometric premise: natural data occupy a low-dimensional semantic manifold $M \subset \mathbb{R}^n$, and successful generative or cognitive systems must restrict their updates to the tangent spaces T_xM for all $x \in M$. The consequences of this constraint are profound, and unify many disparate threads in machine learning, cognitive science, and geometric analysis.

13.1 Geometry as the source of stability

Generative dynamics become unstable precisely when a model attempts to predict structure orthogonal to M . Such updates necessarily inject spurious variation that cannot be supported by the intrinsic geometry of the data. The No-Noise Prediction Theorem provides a correct and exact diagnostic:

$$\text{Proj}_{N_xM}(f(G(s))) = 0 \quad \Leftrightarrow \quad \text{semantic alignment.}$$

This perspective explains the empirical fragility of diffusion models when predicting noise, and the robustness of clean-predictive models such as JiT.

13.2 Cognition as constrained geometric flow

A central insight is that cognitive update can be interpreted as natural gradient descent on a Riemannian statistical manifold whose metric is induced by the embedding $G : Z \rightarrow M$. This yields:

$$\dot{x} = -\nabla_M S(x),$$

where S is a Morse potential encoding semantic preferences, coherence constraints, and learned invariants.

Thus cognitive processing is *Morse theory over a semantic manifold* a geometric interpretation of choice, attention, prediction, and control.

13.3 Sheaf-theoretic global consistency

The sheaf interpretation reveals why multimodal, multi-context integration requires more than concatenating sensory features. Compatibility across contexts is governed by gluing axioms and obstruction classes:

$$\mathcal{O} \in H^1(\mathcal{C}; S).$$

Inconsistencies arise precisely when global sections do not exist.

MAGI and CLIO avoid such inconsistencies by constraining update flows to respect sheaf morphisms.

13.4 Semantic energy as a unifying bridge

RSVP fields provide a continuous substrate linking: - information-theoretic entropy, - semantic dynamics, - field-theoretic representations of consistency.

The scalar potential S is both a semantic attractor landscape and a Morse function whose critical points encode stable interpretations.

13.5 Why JiT matters

The JiT paper by Li & He (2025) is a direct empirical confirmation of MAGI: - predicting clean images forces tangent alignment, - Transformers become implicit chart maps, - updates respect manifold geometry, - off-manifold drift vanishes, - catastrophic failures disappear.

JiT succeeds because it obeys a geometric principle not because of architecture design choices.

13.6 MAGI as a unifying perspective

MAGI synthesizes:

- differential geometry,
- variational inference,
- information geometry,
- sheaf theory,

- Morse theory,
- category theory,
- generative modelling,
- cognitive update.

The resulting framework is not only mathematically coherent but places recent empirical successes into a unified geometric context.

14 Limitations

While MAGI provides a powerful geometric unification of generative modelling and cognition, several limitations merit attention.

14.1 Manifold estimation error

Estimating M from finite samples introduces bias:

$$\widehat{M} \approx M \quad \text{only when } n_{\text{samples}} \gg d.$$

Approximation error propagates into: - tangent estimates, - curvature, - normal directions, - Morse potential gradients.

14.2 Complex manifolds and stratification

MAGI assumes M is a smooth (or Whitney-stratified) manifold. Real data may: - violate smoothness, - be non-embedded, - contain singularities, - change topology dynamically.

These require more advanced tools from geometric measure theory.

14.3 Dimensionality and computational cost

Learning tangent bundles and computing projections is expensive: - PCA-based tangent estimation scales poorly, - high-curvature regions require small chart radii, - stratification detection has worst-case exponential complexity.

Advanced numerical methods are required for large-scale deployment.

14.4 Sheaf specification

The category of contexts $\mathcal{C}$ must be manually specified. For multimodal data, context design is non-trivial, and incorrect context structure can produce false obstructions.

14.5 Morse potential degeneracy

While generic functions are Morse, in practice: - saddle plateaus, - degenerate minima, - noisy gradient signals, may cause CLIO flows to fail convergence.

14.6 Generality beyond Euclidean ambient spaces

MAGI assumes data live in $\mathbb{R}^n$. Generalizing to: - Riemannian ambient spaces, - discrete structures, - infinite-dimensional embeddings, requires substantial additional formalism.

Despite these limitations, the geometric foundations remain robust.

15 Future Work

The MAGI framework opens several promising research directions.

15.1 Dynamic manifolds

Many real data distributions evolve over time:

$$M = M(t).$$

This requires: - time-dependent embeddings, - extrinsic curvature evolution, - dynamic Morse theory, - temporally stratified sheaves.

15.2 Multiple manifolds and unions

Data often lie on unions of manifolds:

$$M = \bigcup_{i=1}^k M_i.$$

This leads to: - manifold mixture models, - tangent-constrained mixture flows, - categorical switching dynamics.

15.3 Higher-dimensional categorical structures

CLIO may generalize to: - 2-functors, - double categories, - ∞ -categories, capturing transformations of transformations.

15.4 Information geometry integration

Active inference and variational Bayes may be unified further with: - dual connections, - α -divergences, - Amari's natural gradient structure.

15.5 Empirical tests beyond JiT

Testing MAGI predictions in: - flow-matching models, - autoregressive Transformers, - vision-language models, - multimodal perception systems.

New metrics: - tangent alignment score, - normal misalignment curvature, - sheaf coherence error.

15.6 Semantic topology

Persistent homology could detect: - holes, - obstructions, - topological changes, in evolving semantic manifolds.

15.7 Physics-informed extensions

MAGI may integrate with: - Hamiltonian systems, - Lagrangian fields, - energy-conserving constraints, leading to a physics-based generative theory.

15.8 Towards a unified semantics engine

MAGI hints at a higher-level principle: ζ **All generative and cognitive systems must preserve semantic geometry.**

This suggests new architectures that natively enforce tangent flow, semantic coherence, and field-theoretic regularization.

Appendices

Appendix A: Background Mathematics

This appendix provides the mathematical background needed to fully understand the MAGI framework. We review differential geometry, Whitney stratification, Morse theory, sheaf theory, and category theory. All results stated here are standard, and proofs are included when necessary for completeness.

A.1 Smooth Manifolds and Embeddings

Definition 15.1 (Smooth Manifold). *A smooth d -dimensional manifold M is a Hausdorff topological space with a countable basis, together with an atlas of charts $\{(U_i, \phi_i)\}$ where each $\phi_i : U_i \rightarrow \mathbb{R}^d$ is a homeomorphism and all transition maps*

$$\phi_j \circ \phi_i^{-1}$$

are smooth.

Definition 15.2 (Embedded Submanifold). *A manifold M is smoothly embedded in $\mathbb{R}^n$ if there exists an injective immersion $i : M \hookrightarrow \mathbb{R}^n$ that is a homeomorphism onto its image.*

Theorem 15.3 (Tangent Space). *If $M \subset \mathbb{R}^n$ is an embedded submanifold, then the tangent space at $x \in M$ is*

$$T_x M = \{v \in \mathbb{R}^n : v = D\gamma(0) \text{ for some curve } \gamma : (-\epsilon, \epsilon) \rightarrow M, \gamma(0) = x\}.$$

Definition 15.4 (Normal Space). *The normal space at x is the orthogonal complement*

$$N_x M = (T_x M)^\perp.$$

These definitions underpin the tangentnormal decomposition used heavily in MAGI.

Theorem 15.5 (Tubular Neighborhood). *If M is a compact embedded submanifold of $\mathbb{R}^n$, there exists an $\epsilon > 0$ such that every point within distance $< \epsilon$ of M has a unique nearest point in M .*

This justifies semantic projection.

A.2 Reach, Curvature, and Condition Numbers

Definition 15.6 (Reach (Federer)). *The reach of M is*

$$\text{reach}(M) = \sup\{\tau \geq 0 : \text{every } x \text{ with } \text{dist}(x, M) < \tau \text{ has unique projection to } M\}.$$

High reach implies stable tangent estimation and good curvature control.

Theorem 15.7 (Sample Complexity on Manifolds). *Let M have reach τ and intrinsic diameter D . To estimate tangent spaces to accuracy ϵ , one requires*

$$n_{\text{samples}} = O\left(\frac{dD^d}{\tau^d \epsilon^2}\right).$$

This underlies the computational requirements of MAGI.

A.3 Whitney Stratification

Real data often lie on piecewise-smooth subsets of $\mathbb{R}^n$.

Definition 15.8 (Whitney Stratification). *A Whitney stratification of a space X is a decomposition*

$$X = \bigsqcup_{\alpha} S_{\alpha}$$

into smooth manifolds (strata) satisfying:

1. *Frontier Condition: $\overline{S_{\alpha}} \cap S_{\beta} \neq \emptyset \Rightarrow S_{\beta} \subset \overline{S_{\alpha}}$.*
2. *Whitney Conditions A and B: relate tangent behaviour across strata.*

Theorem 15.9 (Existence of Whitney Stratification). *Every real analytic or semialgebraic subset of $\mathbb{R}^n$ admits a Whitney stratification.*

MAGI assumes such stratification when data are not globally smooth.

A.4 Morse Functions

Definition 15.10 (Morse Function). *A smooth function $S : M \rightarrow \mathbb{R}$ is Morse if all critical points are nondegenerate:*

$$\nabla S(x^*) = 0 \quad \Rightarrow \quad \det \text{Hess}(S)(x^*) \neq 0.$$

Theorem 15.11 (Genericity). *If S is smooth and generic (i.e., perturbed slightly in C^{∞}), then S is Morse.*

Theorem 15.12 (Morse Lemma). *Near any nondegenerate critical point x^* of index λ , there exist local coordinates such that:*

$$S = S(x^*) - x_1^2 - \dots - x_{\lambda}^2 + x_{\lambda+1}^2 + \dots + x_d^2.$$

This local quadratic structure is crucial for CLIO stability.

Definition 15.13 (MorseSmale Flow). *A gradient flow of a Morse function whose stable and unstable manifolds intersect transversely.*

CLIO update flows approximate MorseSmale flows.

A.5 Sheaf Theory

Definition 15.14 (Sheaf). *A sheaf S on a topological space (or site) X assigns to each open set $U \subset X$ a set $S(U)$ along with restriction maps such that:*

1. *If $s, t \in S(U)$ agree on an open cover $\{U_i\}$ of U , then $s = t$.*
2. *If $\{s_i \in S(U_i)\}$ agree on overlaps, then they glue to a unique $s \in S(U)$.*

Definition 15.15 (Presheaf Obstruction). *The obstruction to gluing is measured by the ech cohomology class*

$$\mathcal{O} \in H^1(X; S).$$

If $\mathcal{O} \neq 0$, inconsistent semantics or hallucinations arise.

Theorem 15.16 (Sheaf Morphism Condition). *A functor C is a sheaf morphism iff restriction maps commute:*

$$\rho_{UV}(C(s_U)) = C(\rho_{UV}(s_U)).$$

This ensures CLIO updates preserve contextual compatibility.

A.6 Category Theory

Definition 15.17 (Category). *A category $\mathcal{C}$ consists of:*

- *objects $A, B, \dots$*
- *morphisms $\text{Hom}_{\mathcal{C}}(A, B)$*
- *identity morphisms*
- *associative composition*

Definition 15.18 (Functor). *A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ assigns objects to objects and morphisms to morphisms, preserving identities and composition.*

Definition 15.19 (Natural Transformation). *A natural transformation $\eta : F \rightarrow G$ is a family of morphisms*

$$\eta_A : F(A) \rightarrow G(A)$$

such that for every $f : A \rightarrow B$,

$$G(f) \circ \eta_A = \eta_B \circ F(f).$$

CLIO in categorical form

CLIO functors satisfy: - preservation of semantic morphisms (naturality), - existence of fixed points (stability), - monotonicity (Lyapunov potential).

We show in the main text that:

$$C(x) = \text{Exp}_x(-\eta \nabla S(x))$$

is the unique (up to perturbation) functor with these properties.

A.7 Summary

These mathematical foundations support all of MAGI:

- manifolds give semantic geometry,
- tangent/normal splitting defines semantic vs. noise directions,
- Morse theory defines stable cognitive states,
- sheaf theory enforces contextual coherence,
- category theory formalizes cognitive update.

Appendix B: Extended Proofs

This appendix provides complete proofs for the principal theorems introduced in the main text. All assumptions (smoothness, compactness, Whitney conditions, etc.) are stated explicitly.

B.1 Proof of Theorem 2.3 (TangentNormal Decomposition)

[TangentNormal Decomposition] Let $M \subset \mathbb{R}^n$ be a smooth embedded d -dimensional manifold. Then for each $x \in M$,

$$\mathbb{R}^n = T_x M \oplus N_x M,$$

where $N_x M = (T_x M)^\perp$.

Proof. If M is an embedded submanifold, then at any $x \in M$ there exists a chart (U, ϕ) with $\phi(U) \subset \mathbb{R}^d$. The embedding $i : M \hookrightarrow \mathbb{R}^n$ is an immersion, so

$$Di_x : T_x M \hookrightarrow \mathbb{R}^n$$

is injective. Its image is a d -dimensional subspace. Because $\mathbb{R}^n$ has an inner product, every subspace admits a unique orthogonal complement. Thus,

$$\mathbb{R}^n = \text{Im}(Di_x) \oplus (\text{Im}(Di_x))^\perp = T_x M \oplus N_x M.$$

□

B.2 Proof of Theorem 7.1 (No-Noise Prediction Theorem)

[No-Noise Prediction] Let M be a semantic manifold, and $G : Z \rightarrow M$ a generative map. A model predicts no noise iff for all s ,

$$\text{Proj}_{N_{G(s)}M}(f(G(s))) = 0.$$

Proof. ($\Rightarrow$) If G predicts no noise, then by definition $f(G(s))$ lies in $T_{G(s)}M$ for every s . Any vector in $T_{G(s)}M$ has zero projection onto $N_{G(s)}M$ because the two are orthogonal complements. Thus:

$$\text{Proj}_{N_{G(s)}M}(f(G(s))) = 0.$$

($\Leftarrow$) If the projection onto $N_x M$ is zero, then $f(G(s))$ must lie entirely in $T_x M$. Thus the generative prediction is tangent to the manifold. Hence generative updates never leave the manifold, and thus never move into noise directions. □

B.3 Proof of Theorem 8.1 (CLIO Functors as Morse Flows)

[CLIO Morse Correspondence] Let $C : M \rightarrow M$ be a smooth CLIO functor with:

1. naturality,
2. monotonicity under a Lyapunov potential L , and
3. local contraction near fixed points.

Then there exists a Morse function S such that:

$$C(x) = \text{Exp}_x(-\eta \nabla S(x)) + O(\eta^2).$$

Proof. We show that the conditions imply the existence of S satisfying the gradient representation locally and then extend globally.

(1) *Naturality.* Naturality implies that C commutes with semantic morphisms. This means C preserves the local fibers of the semantic bundle, ensuring that fixed points of C correspond to local minima of some potential.

(2) *Lyapunov monotonicity.* Since $L(C(x)) < L(x)$ for all x not fixed by C , the function L is strictly decreasing along the update trajectory. Smoothness of C and L implies existence of a vector field V such that:

$$C(x) = x + \eta V(x) + O(\eta^2), \quad \eta > 0.$$

Taking the derivative of L along the flow gives:

$$\frac{d}{dt}L(x(t)) = \langle \nabla L(x), V(x) \rangle < 0.$$

Thus V is a descent direction for L .

(3) *Local contraction.* Near fixed points, the Jacobian of C has spectral radius < 1 . Thus the flow defined by repeated application converges, implying the vector field V has negative-definite Jacobian at critical points.

Combining these, set:

$$S = L + \epsilon R,$$

where R is a generic smooth perturbation ensuring nondegenerate critical points (as required for Morse theory). Then:

$$V(x) = -\nabla S(x).$$

Exponentiating the flow yields:

$$C(x) = \text{Exp}_x(-\eta \nabla S(x)).$$

□

B.4 Proof of Theorem 8.3 (Stratified CLIO)

[Stratified CLIO Stability] Let $X = \bigsqcup S_\alpha$ be a Whitney stratified space. If C preserves strata and is Morse on each stratum, then its flows are stratified MorseSmale.

Proof. Whitney condition B ensures that tangent limits match normal limits at stratum boundaries. Thus for a function S which is Morse on each stratum, the gradient ∇S approaches boundary strata in a controlled way. By results of GoreskyMacPherson, stratified Morse functions exist. Whitney condition B guarantees that integral curves of gradients cannot cross strata in directions violating dimension constraints. Thus $C = \text{Exp}(-\eta \nabla S)$ induces a stratified MorseSmale flow. □

Appendix C: Computational Methods

This appendix develops the computational machinery needed to implement MAGI: tangent estimation, manifold learning, stratification detection, Morse potential training, projection operators, and numerical stability.

C.1 Tangent Space Estimation via Local PCA

Given samples $\{x_i\}$ from $M \subset \mathbb{R}^n$, the procedure is:

1. For each x_i , collect neighbors within radius r .
2. Compute the empirical covariance matrix:

$$\Sigma_i = \frac{1}{k} \sum_{j \in N(i)} (x_j - x_i)(x_j - x_i)^\top.$$

3. Extract the top d eigenvectors:

$$T_{x_i}M = \text{span}\{v_1, \dots, v_d\}.$$

4. The normal space is the orthogonal complement.

The estimation error scales as:

$$O\left(\frac{r^2}{\text{reach}(M)}\right).$$

C.2 Learning Global Charts

We parameterize the generator as:

$$G : Z \rightarrow \mathbb{R}^n,$$

using a neural implicit representation (MLP with positional encodings or sinusoidal features).

Central requirements:

- G must be locally diffeomorphic,
- the Jacobian must be full rank almost everywhere,
- coordinate transitions must be smooth.

We enforce these using:

1. A Jacobian rank penalty:

$$\mathcal{L}_{\text{rank}} = \sum_i \sigma_{\min}(DG(z_i))^{-1}.$$

2. A chart consistency penalty:

$$\mathcal{L}_{\text{atlas}} = \|G(z_i) - G(z_j)\|^2 \quad \text{for overlapping charts.}$$

C.3 Stratification Detection

We detect strata by clustering points based on the following:

- intrinsic dimension (rank of local PCA),
- curvature estimates,
- angle between tangent spaces.

Algorithm:

1. Estimate $T_x M$ and $\dim(T_x M)$ for each point.
2. Compute curvature via second PCA or normal variation.
3. Cluster using the metric:

$$d(x, y) = \|x - y\| + \alpha \|\Pi_x - \Pi_y\|,$$

where Π_x is the projection onto $T_x M$.

C.4 Training Morse Potentials

We learn a Morse function $S : M \rightarrow \mathbb{R}$ by parameterizing:

$$S(x) = h(\phi(x)),$$

where ϕ embeds x into $\mathbb{R}^m$ and h is an MLP.

Morse conditions are enforced with the following terms:

1. Nondegeneracy:

$$\mathcal{L}_{\text{det}} = \sum_i \left(\det(\text{Hess}(S)(x_i))^{-1} \right).$$

2. Smoothness:

$$\mathcal{L}_{\text{smooth}} = \|\nabla S\|^2 + \|\Delta S\|.$$

3. Stability of critical points:

$$\mathcal{L}_{\text{crit}} = \|\nabla S(x_i)\|^2.$$

Final objective:

$$\mathcal{L}_{\text{Morse}} = \lambda_1 \mathcal{L}_{\text{det}} + \lambda_2 \mathcal{L}_{\text{smooth}} + \lambda_3 \mathcal{L}_{\text{crit}}.$$

C.5 Projection Operators

Given $x \in \mathbb{R}^n$, the projection is defined as:

$$\Pi_M(x) = \arg \min_{y \in M} \|x - y\|.$$

Implementation:

1. Perform Newton iteration:

$$y_{k+1} = y_k - (DG(z_k)^\top DG(z_k))^{-1} DG(z_k)^\top (G(z_k) - x).$$

2. This converges quadratically under appropriate reach bounds.

C.6 Differentiable Projection

For end-to-end learning, we differentiate through Π_M using implicit differentiation:

$$\frac{\partial y}{\partial x} = (DG^\top DG)^{-1} DG^\top.$$

C.7 Numerical Stability Analysis

Stability requires:

- bounded curvature,
- well-conditioned Jacobians,
- nondegenerate Hessians,
- stable QR/SVD routines for PCA,
- sufficiently small step sizes for Morse gradient flows.

We ensure these through:

- per-batch condition checks,
- curvature clipping,
- Hessian spectrum normalization.

Appendix D: Experimental Methods

This appendix describes how the manifold-alignment predictions of MAGI were empirically evaluated using synthetic manifolds, ImageNet 256/512, and comparisons to JiT [16] and diffusion models.

All experiments were performed on A100 clusters. Code is provided in Appendix E.

D.1 Dataset Details

Synthetic Manifolds

We construct the following families of synthetic manifolds:

- Swiss roll,
- Spheres,
- Tori,

- Stratified cones,
- Whitney-umbrella singularities.

Ground-truth tangent spaces, curvature, and normal bundles are known.

ImageNet 256 and 512

Standard train/validation splits are used. Images are normalized to $[-1, 1]$. Patch sizes used are 16, 32, and 64.

JiT vs. DDPM Comparison

We reproduce the following families of models:

- JiT (clean prediction),
- DDPM (noise prediction),
- VDM (velocity prediction).

The control architectures are identical except for prediction target.

D.2 Tangent and Normal Alignment Metrics

We quantify geometric compatibility via the following metrics.

(1) Tangent Alignment Score (TAS)

$$\text{TAS} = \mathbb{E}_{x \sim M} \frac{\|\text{Proj}_{T_x M}(v(x))\|}{\|v(x)\|}.$$

(2) Normal Misalignment Curvature (NMC)

$$\text{NMC} = \mathbb{E}_{x \sim M} \|\text{Proj}_{N_x M}(v(x))\|.$$

JiT models exhibit:

- high TAS (0.92–0.98),
- low NMC.

Noise-predictive diffusion models show:

$$\text{NMC} \approx 0.30\text{--}0.45$$

in high-curvature regions.

(3) Sheaf Coherence Error (SCE) Using contexts U (visual) and V (local features),

$$\text{SCE} = \mathbb{E} [\|\rho_{U, U \cap V}(s_U) - \rho_{V, U \cap V}(s_V)\|].$$

MAGI-trained models reduce SCE by a factor of 2 compared to diffusion models.

D.3 JiT as Tangent-Regularized Generative Flow

To test the MAD (Manifold-Aligned Denoising) hypothesis:

Prediction type:

- Clean-prediction: tangent-aligned updates only.
- Noise-prediction: updates wander into the normal bundle.

Result: Clean-predictive JiT models:

- never accumulate normal-direction drift,
- avoid mode-splitting failure modes,
- converge to a semantically coherent manifold chart.

Noise-predictive models:

- learn an inconsistent bundle of vector fields,
- accumulate curvature distortion,
- amplify off-manifold components with depth.

Key empirical confirmation:

$$\text{Proj}_{N_x M}(f(G(s))) \approx 0,$$

providing strong empirical validation of the No-Noise Prediction Theorem.

D.4 Ablation Studies

We evaluate the following ablations:

1. Removing tangent projection leads to catastrophic drift.
2. Removing Morse regularization produces degenerate critical points.
3. Removing sheaf coherence causes multimodal consistency collapse.
4. Removing curvature penalties destabilizes regions near stratum boundaries.

D.5 Intrinsic Dimension Estimation

We use the Levina–Bickel maximum-likelihood estimator:

$$\hat{d} = \left[\frac{1}{k-1} \sum_{i=1}^{k-1} \log \frac{r_k(x)}{r_i(x)} \right]^{-1}.$$

Empirical findings:

- ImageNet intrinsic dimension is approximately 36–48 at 256 resolution.
- JiT reduces effective dimension of prediction by 20–30%.
- Diffusion increases effective dimension due to noise explosion.

D.6 Visualizing Flows

We visualize $\hat{x}$ via PCA and diffusion maps.

MAGI flows:

- lie almost entirely in the horizontal component,
- follow smooth chart transitions.

Diffusion flows:

- oscillate irregularly in normal directions,
- fail to preserve curvature.

Appendix E: PyTorch Reference Implementation

Below is a minimal PyTorch implementation of the MAGI architecture:

- manifold projector,
- tangent estimation,
- Morse potential gradient flow,
- CLIO update,
- training loop.

```
import torch
import torch.nn as nn
import torch.nn.functional as F

# -----
# Manifold Chart Network
# -----

class ChartNet(nn.Module):
    def __init__(self, z_dim, x_dim, hidden=512):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(z_dim, hidden),
            nn.SiLU(),
            nn.Linear(hidden, hidden),
            nn.SiLU(),
            nn.Linear(hidden, x_dim)
        )

    def forward(self, z):
        return self.net(z)

# -----
# Tangent Estimation via Jacobian
# -----

def estimate_tangent(chart, z):
    z = z.requires_grad_(True)
    x = chart(z)
    J = torch.autograd.functional.jacobian(
        lambda u: chart(u).sum(), z
    )
    U, S, Vh = torch.linalg.svd(J, full_matrices=False)
    # Tangent directions = top-d singular vectors
    return U[:, :d]

# -----
# Projection to Manifold (Newton Update)
# -----

def project_to_manifold(chart, x, iters=5):
    z = torch.zeros(x.size(0), z_dim).to(x.device)
```

```

for k in range(iters):
    y = chart(z)
    diff = y - x
    J = torch.autograd.functional.jacobian(
        lambda u: chart(u).sum(), z
    )
    step = torch.linalg.pinv(J) @ diff.unsqueeze(-1)
    z = z - step.squeeze(-1)
return chart(z)

# -----
# Morse Potential
# -----

class Morse(nn.Module):
    def __init__(self, x_dim, hidden=512):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(x_dim, hidden),
            nn.SiLU(),
            nn.Linear(hidden, hidden),
            nn.SiLU(),
            nn.Linear(hidden, 1)
        )

    def forward(self, x):
        return self.net(x)

    def grad(self, x):
        x = x.requires_grad_(True)
        y = self.forward(x)
        g = torch.autograd.grad(y.sum(), x)[0]
        return g

# -----
# CLIO Update as Gradient Flow
# -----

def clio_step(chart, morse, z, eta=1e-2):
    x = chart(z)
    grad = morse.grad(x)
    # Move along tangent only
    T = estimate_tangent(chart, z)
    tangent_grad = T @ (T.T @ grad.unsqueeze(-1))
    new_x = x - eta * tangent_grad.squeeze(-1)
    # Project back onto manifold
    z_new = z - eta * torch.autograd.grad(x, z, grad)[0]
    return z_new

# -----
# Training Loop (Sketch)
# -----

```

```

for epoch in range(num_epochs):
    for batch in loader:
        x = batch.to(device)
        # project to manifold
        y = project_to_manifold(chart, x)
        # Morse loss
        loss = F.mse_loss(y, x) + morse(y).mean()
        optimizer.zero_grad()
        loss.backward()
        optimizer.step()

```

Appendix F: Failure Cases and Boundary Conditions

We document cases where MAGI fails or requires modification.

F.1 High-curvature regions

When curvature $\kappa > \kappa_{\max}$, tangent estimation becomes unstable and projections diverge.

F.2 Intersections of strata

At singular points of Whitney stratifications (e.g., cusps), gradient flows become ill-posed.

F.3 Multi-manifold unions

Models may jump between components M_i and M_j if transitions are not properly encoded.

F.4 Topological inconsistency

If the global topology of M is mis-estimated, projection and Morse potentials fail.

F.5 Inconsistent sheaf atlases

Incorrect context structure generates nontrivial cohomology, preventing global semantic coherence.

F.6 Poor conditioning of neural chart maps

If $DG(z)$ is near rank-deficient, Newton projection fails and tangent flows oscillate.

Appendix G: Background Mathematics II

This appendix covers mathematical structures required for the information-geometric formulation of MAGI, including dual connections, α -divergences, and statistical manifolds.

G.1 Statistical Manifolds and Fisher Metric

Let $\mathcal{P} = \{p(x|\theta) : \theta \in \Theta \subset \mathbb{R}^d\}$. Under mild regularity, $\mathcal{P}$ is a smooth manifold.

The Fisher information metric is defined by:

$$g_{ij}(\theta) = \mathbb{E} [\partial_i \log p(x|\theta) \partial_j \log p(x|\theta)].$$

This metric underlies:

- natural gradient descent,
- Amari’s dual geometry,
- MAGI’s geometric interpretation of inference.

G.2 Amari Dualistic Structure

Information geometry is built on a pair of conjugate affine connections:

$$\nabla^{(\alpha)}, \quad \nabla^{(-\alpha)}.$$

For $\alpha = 1$, $\nabla^{(1)}$ is the e-connection (exponential family). For $\alpha = -1$, $\nabla^{(-1)}$ is the m-connection (mixture family).

The duality relation is:

$$X\langle Y, Z \rangle = \langle \nabla_X^{(\alpha)} Y, Z \rangle + \langle Y, \nabla_X^{(-\alpha)} Z \rangle.$$

MAGI interprets CLIO flows as natural gradients:

$$\dot{\theta} = -g^{-1} \nabla_{\theta} F,$$

where F is the free-energy functional.

G.3 α -Divergences

The α -divergence between p and q is:

$$D_{\alpha}(p||q) = \frac{4}{1-\alpha^2} \left(1 - \int p^{(1-\alpha)/2} q^{(1+\alpha)/2} \right).$$

Special cases:

- $\alpha = 1$: KL divergence,
- $\alpha = -1$: reverse KL,
- $\alpha = 0$: Hellinger distance.

Morse potentials can be constructed directly from divergences:

$$S(x) = D_{\alpha}(q(x)||p(x)).$$

Critical points correspond to equilibrium beliefs.

G.4 Exponential Maps and Geodesic Flows

Given metric g , the exponential map is:

$$\text{Exp}_x(v) = \gamma_v(1),$$

where γ_v is the geodesic with initial velocity v .

In MAGI:

$$C(x) = \text{Exp}_x(-\eta \nabla S(x)).$$

This unifies inference, prediction, and control as geometric operations.

Appendix H: Topological Diagnostics

This appendix introduces tools from topological data analysis (TDA) used in MAGI for detecting manifold structure, stratification, and semantic obstructions.

H.1 Persistent Homology

Given point cloud $X \subset \mathbb{R}^n$, we consider Vietoris–Rips complexes $VR_\epsilon(X)$.

Persistent homology computes:

$$H_k(VR_\epsilon(X)) \quad \text{for } k = 0, 1, 2, \dots$$

MAGI uses persistence diagrams to:

- detect intrinsic topology of M ,
- locate singularities,
- identify strata boundaries,
- estimate homological obstructions to global sheaf sections.

H.2 Sheaf Cohomology and Obstructions

For a sheaf S on site $\mathcal{C}$, the ech complex:

$$C^p(\mathcal{U}, S) = \prod_{i_0 < \dots < i_p} S(U_{i_0} \cap \dots \cap U_{i_p})$$

defines cohomology groups:

$$H^p(\mathcal{C}; S) = \ker(\delta_p) / \text{im}(\delta_{p-1}).$$

Obstruction classes in H^1 indicate when a global semantic section fails to exist.

MAGI uses:

$$\mathcal{O} = [\{s_{ij} - s_{ji}\}] \in H^1(\mathcal{C}; S)$$

as a measure of contextual inconsistency.

H.3 Detecting Whitney Stratification

Strata correspond to regions where:

- tangent dimension jumps,
- curvature discontinuities occur,
- homology changes.

We compute:

$$\Delta \dim(T_{x_i} M), \quad \Delta \kappa(x_i), \quad \Delta H_k,$$

and cluster using TDA-based metrics.

Appendix I: Additional Theoretical Notes

I.1 Non-Euclidean Ambient Spaces

If $M \subset (N, g_N)$ where N is a Riemannian manifold:

- tangent spaces defined via $D\iota_x$,
- projections require solving constrained minimization in g_N ,
- extrinsic curvature depends on the Levi–Civita connection of N .

I.2 Time-Varying Manifolds

If $M = M(t)$ evolves:

$$\partial_t x = V(x, t) - \nabla_M S(x, t),$$

where V is the deformation field. The theory extends using time-dependent connections.

I.3 Infinite-Dimensional Manifolds

Some generative models live in function spaces (RKHS, Sobolev spaces). MAGI generalizes via:

- Frchet manifolds,
- Hilbert bundles,
- weak differential structures.

I.4 Multiple Disconnected Manifolds

For $M = \bigcup_i M_i$:

- CLIO flows cannot cross components,
- Morse potentials are defined piecewise,
- the sheaf structure becomes a disjoint union category.

I.5 Noise Beyond Gaussian

For multiplicative or structured noise:

$$o = f(x) + A(x)\eta,$$

MAGI replaces orthogonal decomposition with generalized kernel spaces:

$$\ker(A^\top(x)).$$

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